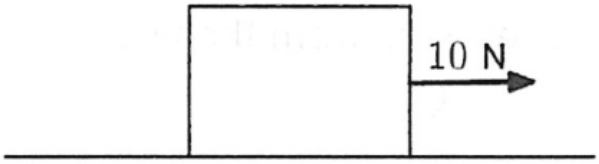
Forces: Coefficient of friction

Questions 1–26

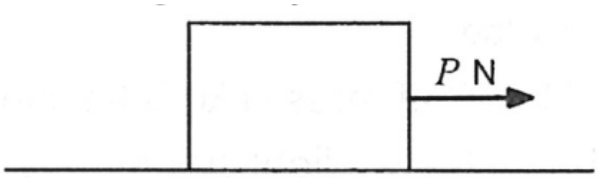
Question 1. A particle of weight w N lies on a rough horizontal surface. A horizontal force 10 N is applied to the particle. The coefficient of friction between the particle and the surface is given by μ



The frictional force F N is the only resistance force, and the particle is at the point of moving. Find the value of F given

a) $w = 50, \mu = 0.2$
b) $w = 20, \mu = 0.45$
c) $w = 15, \mu = 0.85$
d) $w = 3, \mu = 1.1$

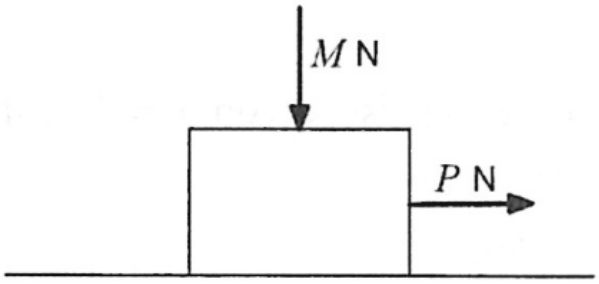
Question 2. A crate of weight w N lies on a rough horizontal surface. A horizontal force P N is applied to the crate. The frictional force F N is the only resistance against the crate. The coefficient of friction between the crate and the surface is given by μ



Given the crate is at the point of moving and

a) $w = 12, \mu = 0.5$, find the value of P
b) $w = 12, \mu = 0.9$, find the value of P
c) $P = 20, \mu = 0.4$, find the value of w
d) $P = 20, \mu = 0.8$, find the value of w
e) $w = 50, P = 40$, find the value of μ
f) $w = 300, P = 12$, find the value of μ
g) $w = 12, P = 15$, find the value of μ

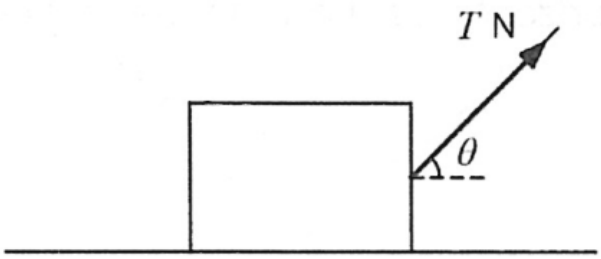
Question 2 (continued). The crate has objects added to it with weight M N as shown in the diagram below. The other forces on the crate remain the same.



Given the crate is at the point of moving and

h) $w = 12, \mu = 0.7, M = 3$, find the value of P
i) $P = 8, \mu = 0.8, M = 2$, find the value of w
j) $w = 60, P = 10, M = 20$ find the value of μ

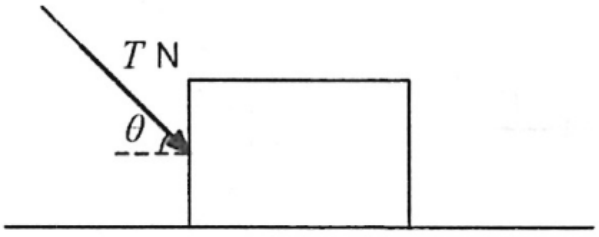
Question 3. A box of weight w N lies on a rough horizontal surface. It is being pulled by a horizontal force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

a) $w = 20, T = 15, \theta = 30, \mu = 0.5$, find the value of F
b) $w = 8, \theta = 45, \mu = 0.2$, find the value of T
c) $T = 10, \theta = 60, \mu = 0.1$, find the value of w
d) $w = 15, T = 4, \theta = 30$, find the value of μ

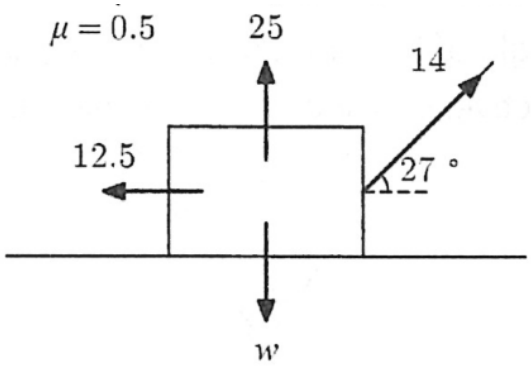
Question 4. A box of weight w N lies on a rough horizontal surface. It is being pushed by a force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

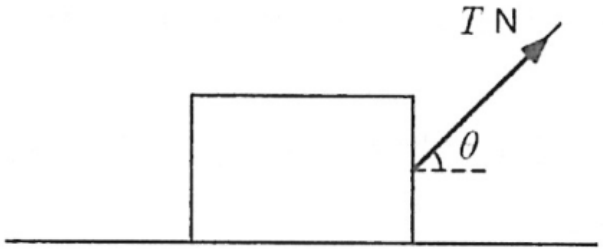
a) $w = 2, T = 4, \theta = 30, \mu = \frac{1}{2}\sqrt{3}$, find the value of F
b) $w = 4, \theta = 45, \mu = 0.3$, find the value of T
c) $T = 12, \theta = 60, \mu = 0.2$, find the value of w
d) $w = 20, T = 5, \theta = 30$, find the value of μ

Question 5. Will has drawn a correct force diagram for a particle in equilibrium, and some workings:



$R(\uparrow)$: $25 + 14\sin 27° - w = 0 \therefore$ mass is 31.4 kg
Explain why his claim that the mass of the particle is 31.4 kg is incorrect.

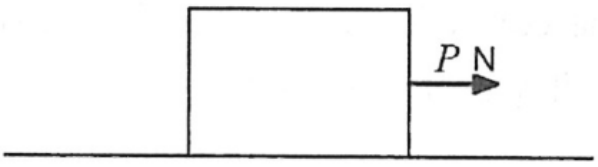
Question 6. The following questions use $g = 9.8$ ms^{-2} A box of mass m kg lies on a rough horizontal surface. It is being pulled by a force T N at an angle $\theta°$ to the horizontal. The frictional force F N is the only resistance against the box. The coefficient of friction between the box and the surface is given by μ



Given the box is at the point of moving and

a) $m = 5, \theta = 10, \mu = 0.5$, find the value of T
b) $T = 20, \theta = 50, \mu = 0.1$, find the value of m
c) $m = 2, T = 5, \theta = 70$, find the value of μ

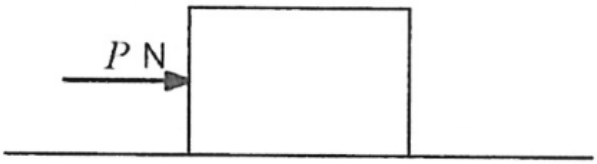
Question 7. The coefficient of friction between a particle of mass 2 kg, on a rough horizontal surface is 0.4. A horizontal force of P N acts on the particle.



In each case, identify if the particle is moving under the given value of P and whether, when an additional mass of 100 g is added to the particle, it continues to move or remain stationary.

a) $P = 8.4$
b) $P = 8$
c) $P = 7.6$

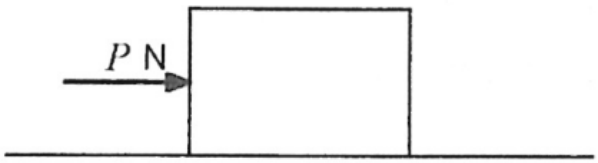
Question 8. A crate of mass m kg lies on a rough floor. The coefficient of friction between the crate and the floor is 0.3. The crate is pushed by a horizontal force P N and it does not move.



For each of the following, find the greatest possible values of P

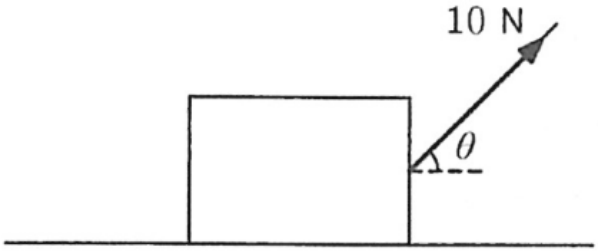
a) $m = 5$ kg
b) $m = 10$ kg
c) $m = 20$ kg

Question 9. The following questions use exact values of g A box of mass m kg lies on a rough floor. The coefficient of friction between the crate and the floor is μ. The crate is pushed by a horizontal force P N and it does not move.



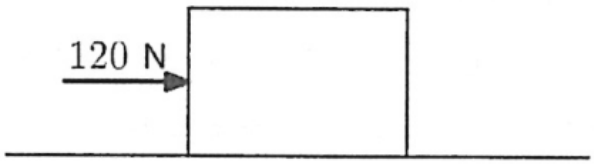
Show that $P \leq \mu mg$

Question 10. The following questions use $g = 10\ \text{ms}^{-2}$ A block of mass 15 kg lies on a rough horizontal surface. The block is being pulled by a force of 10 N at an angle $\theta°$ to the horizontal. The coefficient of friction between the block and the mass is 0.1



Given the block is moving, show that $10\cos\theta + \sin\theta > 15$

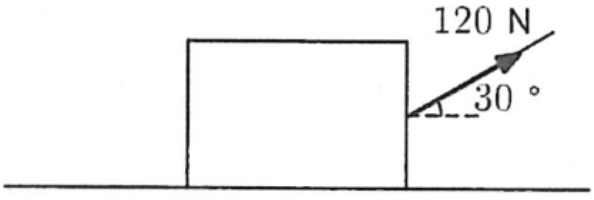
Question 11. A block of mass 15 kg lies on a rough horizontal surface. The coefficient of friction between the block and the mass is 0.85



The block is pushed by a horizontal force of 120 N.

a) Does the block move, and if so, what is it acceleration?

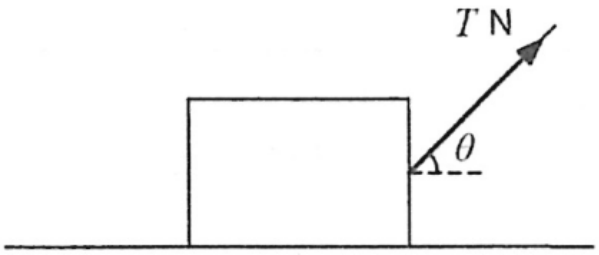
The pushing force is removed, and instead a pulling force of 120 N at angle of 30° to the horizontal is applied.



b) Find the acceleration of the block.

c) Find the maximum extra mass m kg which may be added to the block, for it still to be moving.

Question 12. The following questions use exact values of g A box of mass m kg lies on a rough horizontal surface. It is being pulled by a horizontal force T N at an angle $\theta°$ to the horizontal. The coefficient of friction between the box and the surface is μ. The frictional force F N is the only resistance against the box.

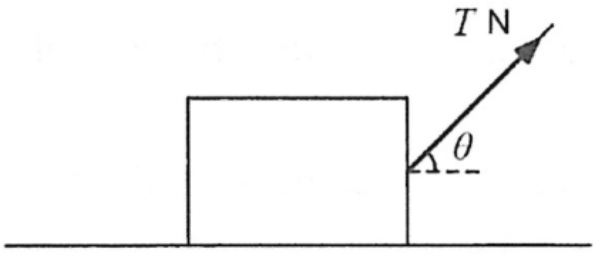


Show that $T \geq \frac{\mu mg}{\cos\theta + \mu \sin\theta}$

Question 13. For each of the following, find the exact values of $\sin\theta$ and $\cos\theta$:

a) $\tan\theta = \frac{3}{4}$
b) $\tan\theta = \frac{5}{12}$
c) $\tan\theta = \frac{7}{16}$
d) $\tan\theta = \frac{10}{21}$

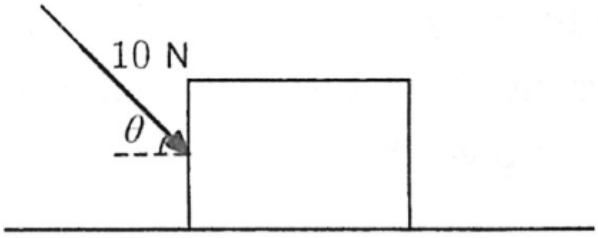
Question 14. The following questions use $g = 9.8$ ms^{-2} A box of mass m kg lies on a rough horizontal surface. It is being pulled by a force T N at an angle $\theta°$ to the horizontal where $\tan\theta = \frac{3}{4}$.
The coefficient of friction between the box and the surface is μ. The frictional force F N is the only resistance against the box.



Given the box is moving and the acceleration of the box is a ms^{-2}, and

a) $m = 10, T = 20, \mu = 0.1$, find the value of a
b) $m = 3, T = 20, \mu = 0.2$, find the value of a
c) $m = 5, T = 15, a = 1.5$, find the value of μ
d) $T = 16, \mu = 0.25, a = 3$, find the value of m
e) in general, show that $a = (\frac{4}{5m} + \frac{3}{5m}\mu)T - 9.8\mu$

Question 15. A box of mass 4 kg lies on a rough horizontal surface. It is being pushed by a force 10 N at an angle $\theta°$ to the horizontal where $\tan\theta = \frac{5}{12}$
The coefficient of friction between the box and the surface is 0.2.



a) Find the acceleration of the box.

The box has additional mass of m kg added to it, given

b) $m = 0.1$ find the value of a
c) $m = 0.5$ find the value of a

Question 16. A trolley of mass 8 kg and is being pushed with a horizontal force of P N along a rough horizontal floor. It is at the point of moving.
In doubling the horizontal pushing force, the trolley moves with acceleration 0.25 ms^{-2}. Find

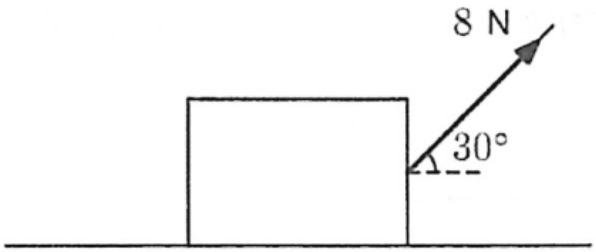
a) the value of P

b) the coefficient of friction between the trolley and the floor.

The trolley is loaded with an extra m kg of products. The trolley is pushed at $2P$ N.

c) Find the maximum value of m such that the trolley is moving or on the point of moving.

Question 17. A sledge of mass 1.2 kg lies on a rough horizontal ground and is being pulled by a force 8 N at an angle 30° to the horizontal, having initially started at rest. The coefficient of friction between the sledge and the ground is 0.2

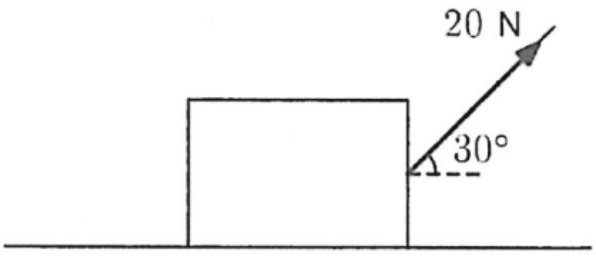


a) Find the acceleration of the sledge.

After 5 seconds, the rope snaps. The sledge continues to move until it comes to a stop.

b) Find the distance the sledge travels after the rope breaks.

Question 18. The following questions use $g = 9.81$ ms^{-2} A crate of mass 5 kg lies on a rough horizontal surface. It is being pulled by a force 20 N at an angle 30° to the horizontal. The coefficient of friction between the crate and the surface is 0.4

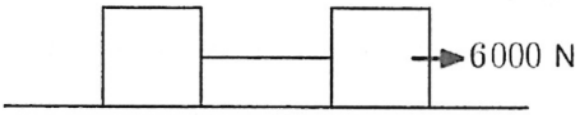


a) Find the acceleration of the crate.

The 20 N force needs to be replaced by a horizontal pushing force P N such that it moves at the same acceleration.

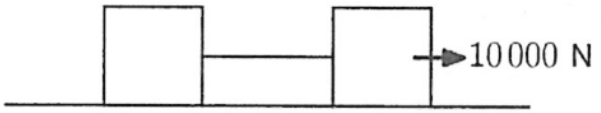
b) Find the value of P

Question 19. A car of mass 2000 kg is towing a trailer of mass 500 kg along a straight flat road. The driving force produced by the engine is 6000 N. The coefficient of friction between the road and the vehicles is 0.1



a) Draw a force diagram.
b) Find the acceleration of the car.
c) Find the tension in the tow bar.
d) State any modelling assumptions you have made.

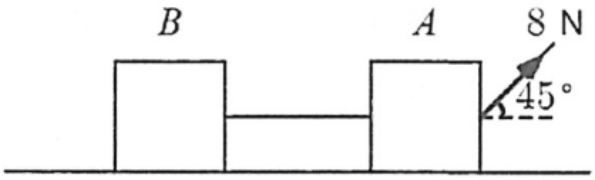
Question 20. A van of mass 3500 kg is towing a trailer of mass 750 kg along a straight flat road. The driving force produced by the engine is 10000 N. The tow bar is modelled as both light and inextensible.



The van is accelerating at 1.5 ms^{-2}.
The coefficient of friction between both the van and the road, and the trailer and the road is μ

a) Find the value of μ
b) Find the tension in the tow bar.
c) Explain how the modelling assumptions that the string is light and inextensible have been used.

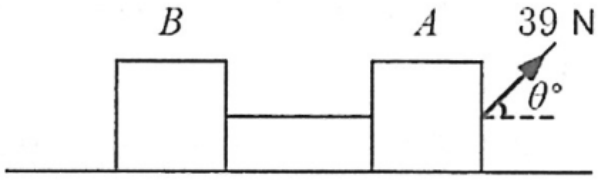
Question 21. The following questions use $g = 9.8$ ms^{-2} A toy train is made up of an engine A of mass 0.8 kg connected by a horizontal, light, inextensible string to truck B of mass 0.3 kg.
The engine is being pulled by a light inextensible string with tension 8 N which is at an angle of 45° to the horizontal.



The coefficient of friction between both the engine and the truck and the floor is 0.95

a) Draw a force diagram.
b) Find the friction force acting on the engine.
c) Find the friction force acting on the truck.
d) Find the acceleration of the engine.
e) Find the tension in the string connecting the engine and the truck.
f) State a modelling assumption you have used.

Question 22. A particle A of mass 10 kg and a particle B of mass 5 kg are on a horizontal rough surface and are connected by a light, inextensible string which remains taut.
Particle A is being pulled by a light, inextensible string, at an angle of $\theta°$ where $\tan\theta = \frac{5}{12}$. The tension in this string is 39 N.



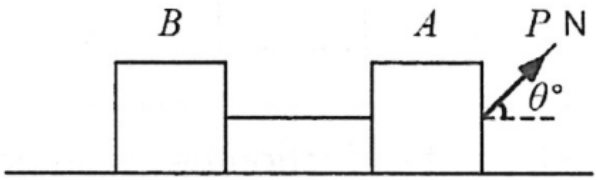
The coefficient of friction between the particles and the surface is μ.
The deceleration of the particles is 3.5 ms^{-2}.

a) Draw a force diagram.
b) Find the coefficient of friction between the particles and the surface.
c) Find the tension in the string connecting the two particles.
d) State a modelling assumption that you have used.

It takes 10 seconds before coming to rest.

e) Find the distance they have travelled.

Question 23. The following questions use $g = 10$ ms^{-2} A particle A of mass 8 kg and a particle B of mass 3 kg are on a horizontal rough surface and are connected by a light, horizontal, inextensible string which remains taut.
Particle A is being pulled by a light, inextensible string, at an angle of $\theta°$ where $\tan\theta = \frac{7}{24}$. The tension in this string is P N.



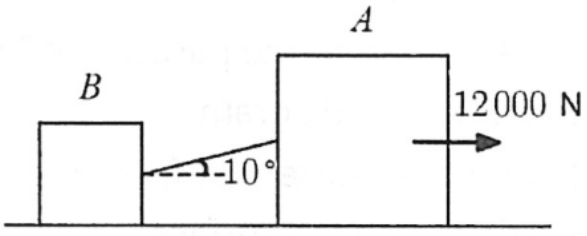
The coefficient of friction between the particles and the surface is μ.
The system is initially at rest, then the acceleration of the particles is 2.5 ms^{-2}.

a) Draw a force diagram.

It is given that the friction acting against the motion of particle B is 6 N. Find

b) the value of μ
c) the tension in the string connecting particle A and the particle B.
d) the value of P.
e) the friction acting against particle A.
f) How long it takes for the particles to travel 20 metres.

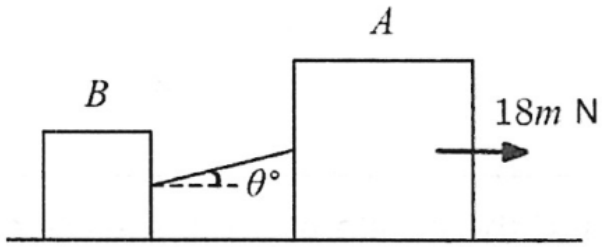
Question 24. A van of mass 4000 kg is towing a trailer B of mass 1000 kg along a straight flat road. The driving force produced by the engine is 12000 N.
The tow bar is angled upwards from B to A with angle 10° to the horizontal. The tow bar may be modelled as a rod.



The coefficient of friction between the van and trailer and the surface is μ
The van is accelerating at 0.5 ms^{-2}.

a) Draw a force diagram.
b) Find the value of μ
c) Find the tension in the tow bar.

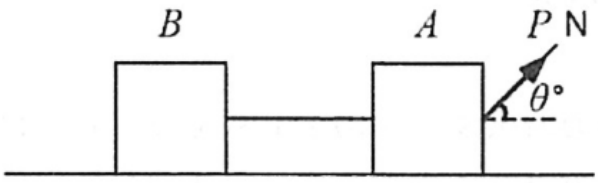
Question 25. A particle A of mass m kg and a particle B of mass $3m$ kg are on a horizontal rough surface and are connected by a light, horizontal, inextensible string which remains taut. Particle A is pulled forward by a force of $18m$ N.
The tow bar is angled upwards from B to A with angle θ° to the horizontal. The tow bar may be modelled as a rod.



The coefficient of friction between the van and trailer and the surface is μ
The van is accelerating at 2 ms^{-2}.

a) Show that $\mu = 0.25$
b) Show that $T = \frac{54m}{\sin\theta + 4\cos\theta}$

Question 26. The following questions use exact values of g A particle A of mass $4m$ kg and a particle B of mass m kg are on a horizontal rough surface and are connected by a light, inextensible string. Particle A is being pulled by a force of P N, at an angle of $\theta°$ to the horizontal where $\tan\theta = \frac{3}{4}$. The coefficient of friction between the particles and the surface is μ.



a) Draw a force diagram.

Given that $P = 25$ and the total resistance force is 10 N.

b) Show that $\mu = \frac{2}{mg-3}$

c) Show that the acceleration of the system is given by $a = \frac{2}{m}$ ms^{-2}

d) Find the tension in the string between the two particles in terms of m